Mae **Subramanian**

https://mae.red
ms7832@rit.edu

(646) 864-7353
Rochester, NY

Engineering Portfolio

Let's work together and build something incredible.
Visit https://mae.red/projects/ for an interactive version.

Contents

Machine Room Security Kiosk **2**
- Exploded View 3
- Card Unlocker 3

Cobot-Automated PCBA Test Fixture **4**
- Fixture Internals 6
- Test Cart & System Design 7

IoT-Enabled Landscape Lighting *(Done Right)* **8**
- Problem Statement 8
- Drafting the Installation 8
- Control System 9
- Project Gallery: 10

Machine Room Security Kiosk

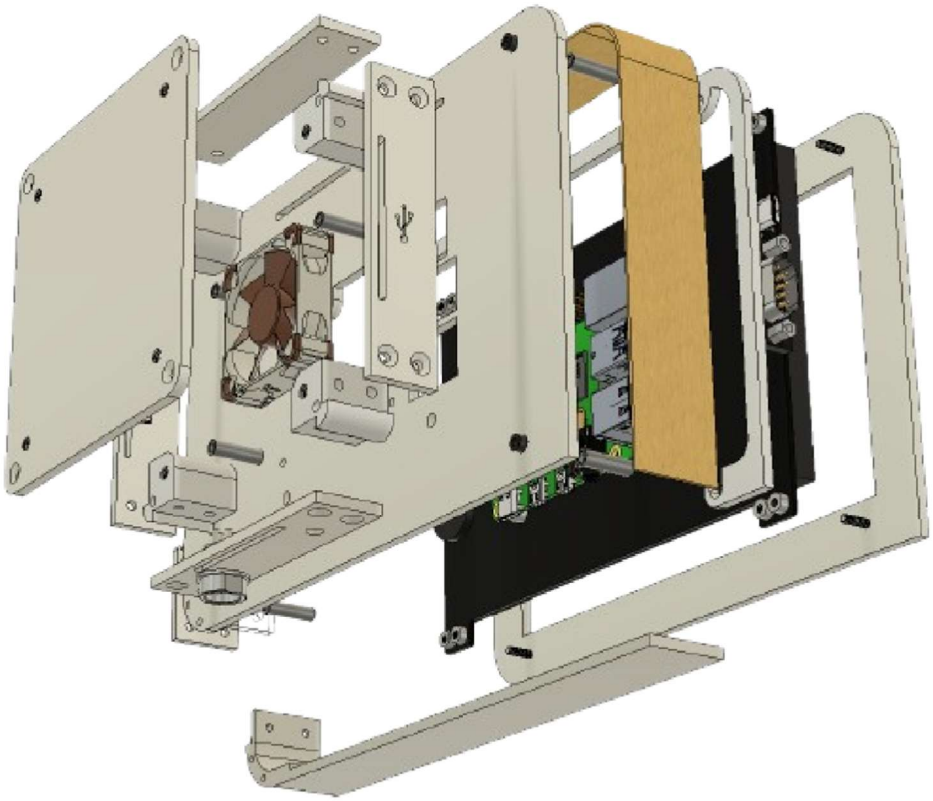
→ See it in action on YouTube

A complete identity management and security system that sits outside of the machine room at my former high school. It lets students in by PIN number and tracking which machines they use. *As of September 2026, this system has been used* ***over 18 thousand times.***



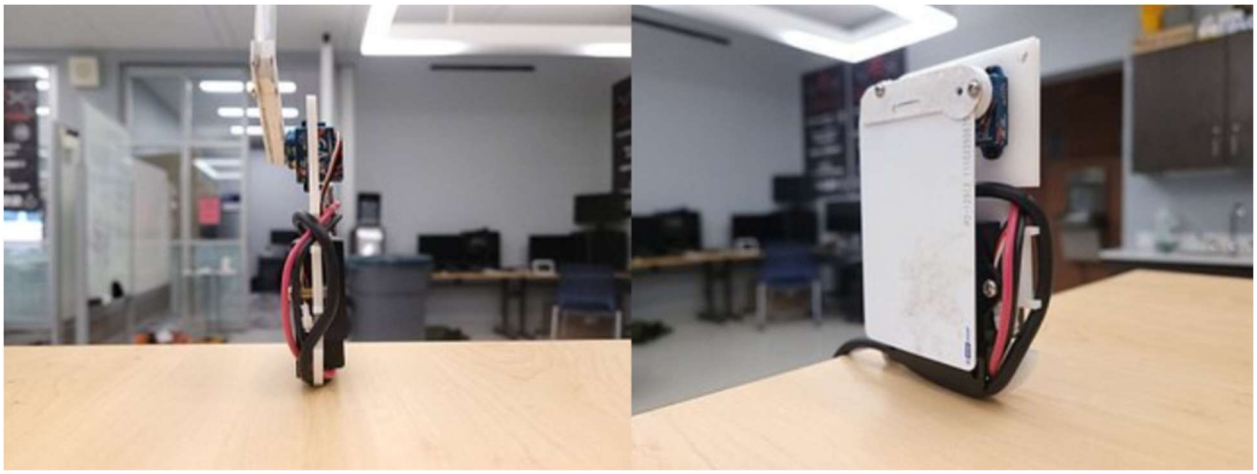
Exploded View

The kiosk body was built using laser-cut Delrin, 3D printed parts, and a custom-bent gold-anodized aluminum trim.



Card Unlocker

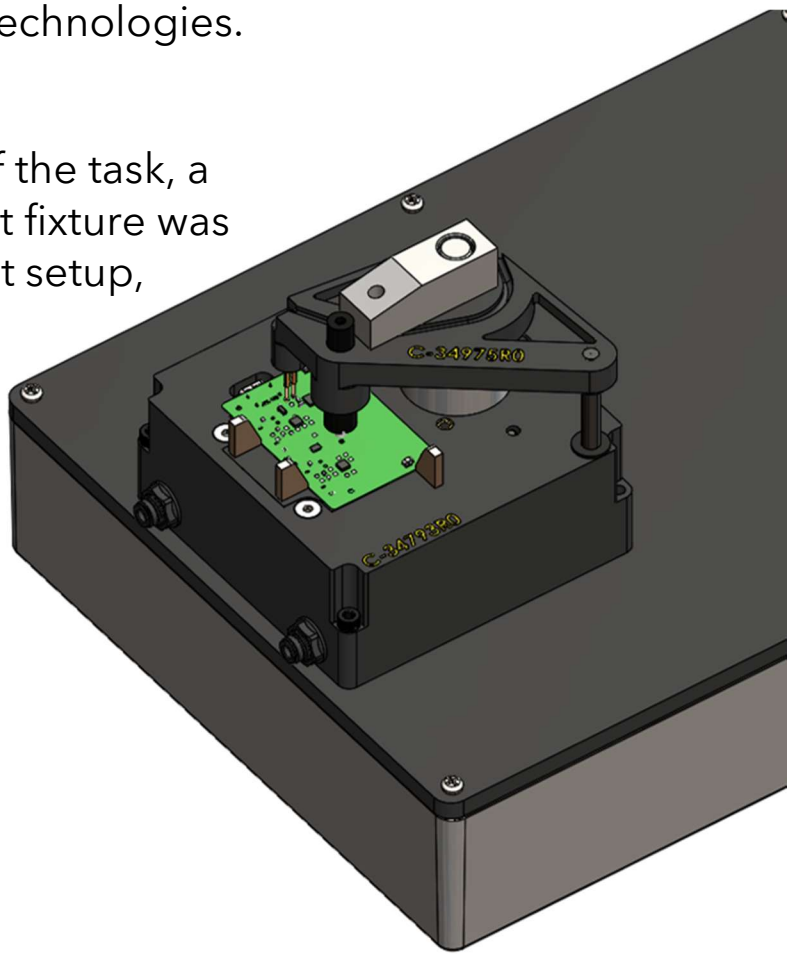
The door unlocker or "card swiper" takes the place of an exhausted computer science teacher tasked with unlocking the room multiple times a day for robotics students. MHS security requires the use of a supplied RFID keycard for the machine shop door. Within a split-second, the unlocker briefly actuates the keycard in front of the reader.

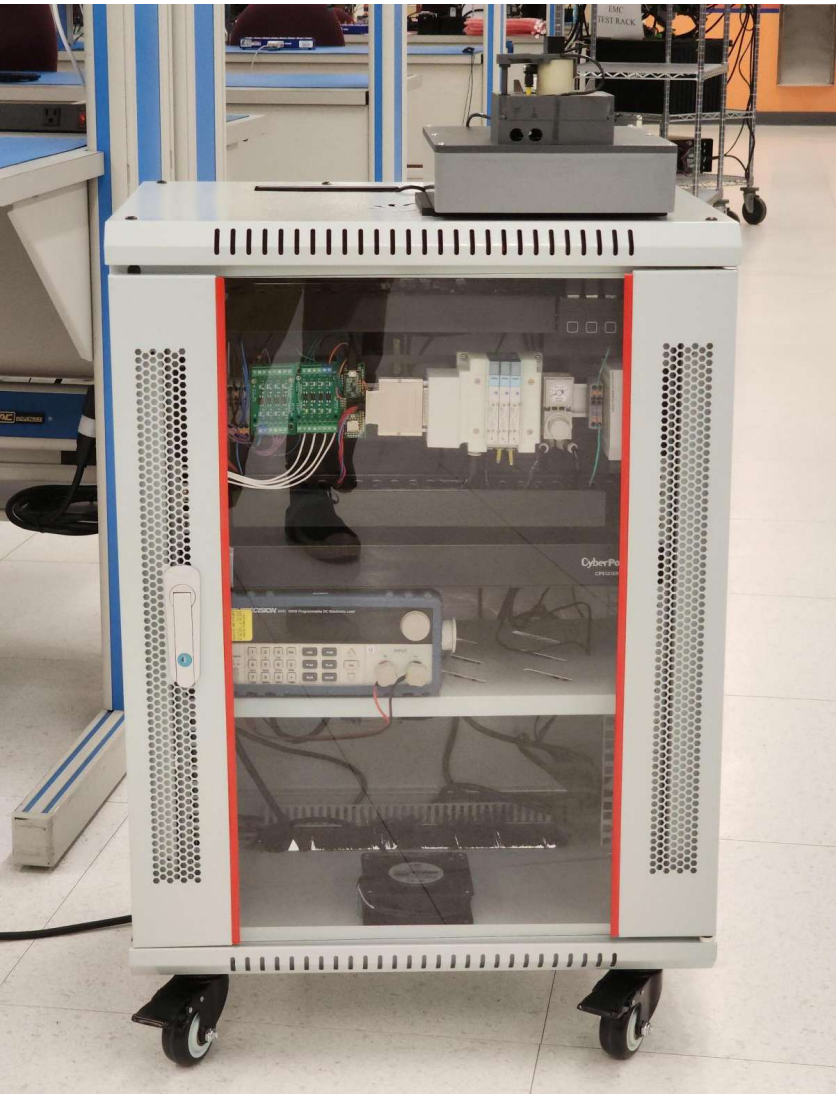


Cobot-Automated PCBA Test Fixture

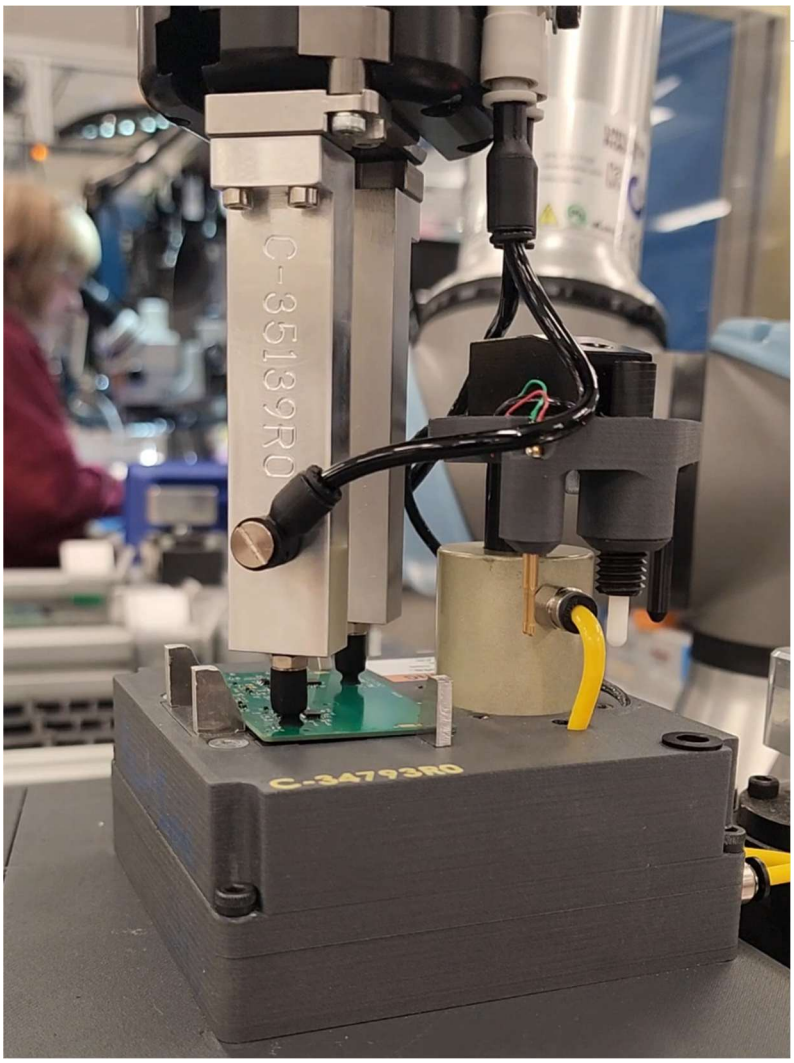
Manual “pick, place, start” tasks caused significant slowdowns on the PCBA production floor at Enercon Technologies.

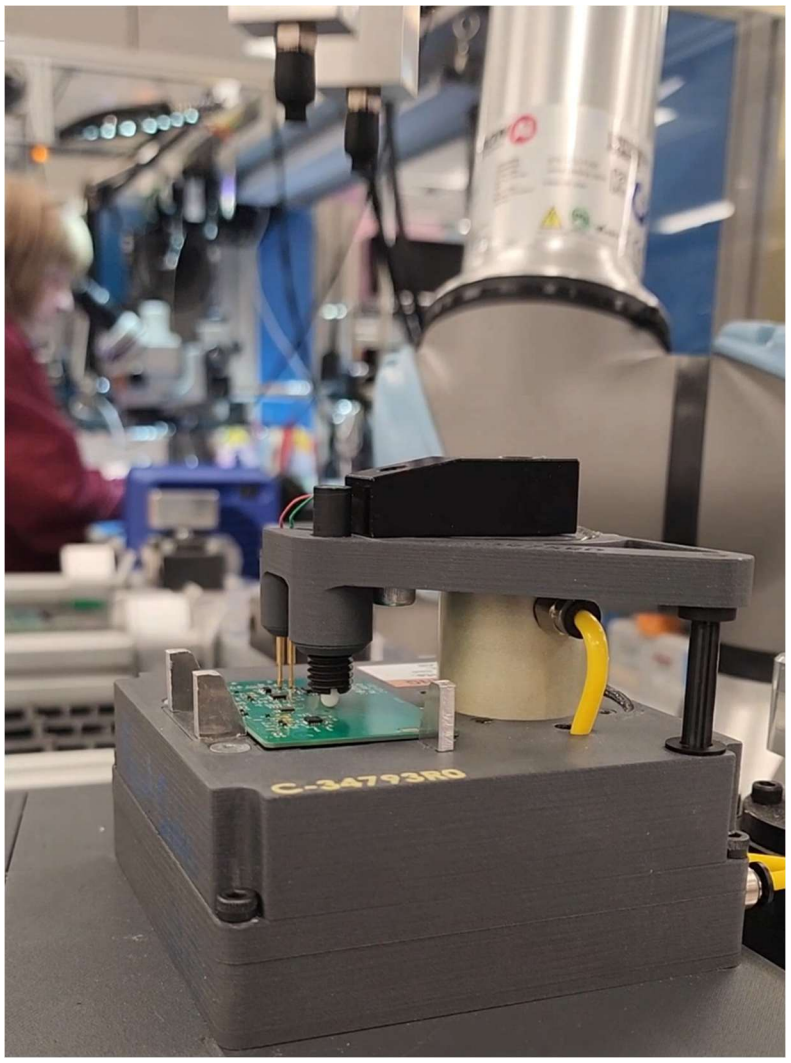
To allow for complete automation of the task, a cobot-optimized rendition of the test fixture was created along with a complete cobot setup, programming, and testing platform.





An all-in-one, **modular** mobile test station is able to accommodate up to 4 fixtures at once. Once placed by the cobot, a pneumatic swing clamp secures the board and powers it via two test points. The completed system can accommodate for over **500 boards per day**, as opposed to the average 300/day throughput.





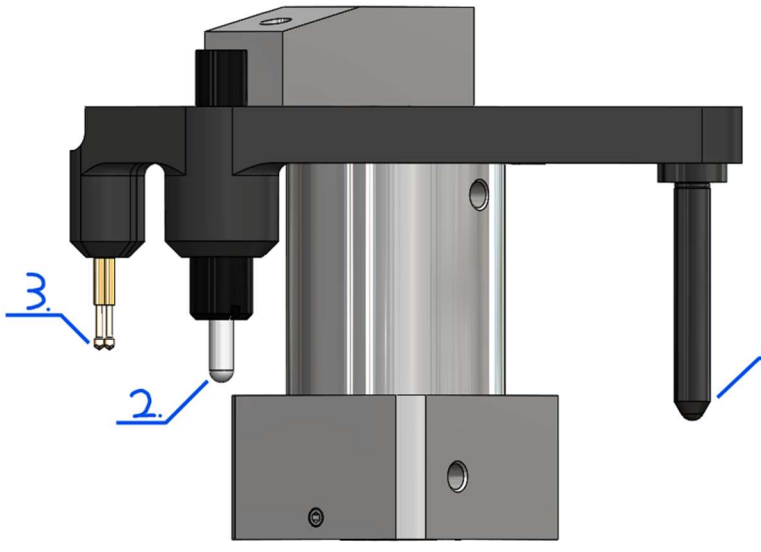
Pneumatic Swing Clamp

- Clears vertical space above DUT in resting state
- Establishes electrical connection

Actuation within **250ms**

Swing Clamp Feature Diagram

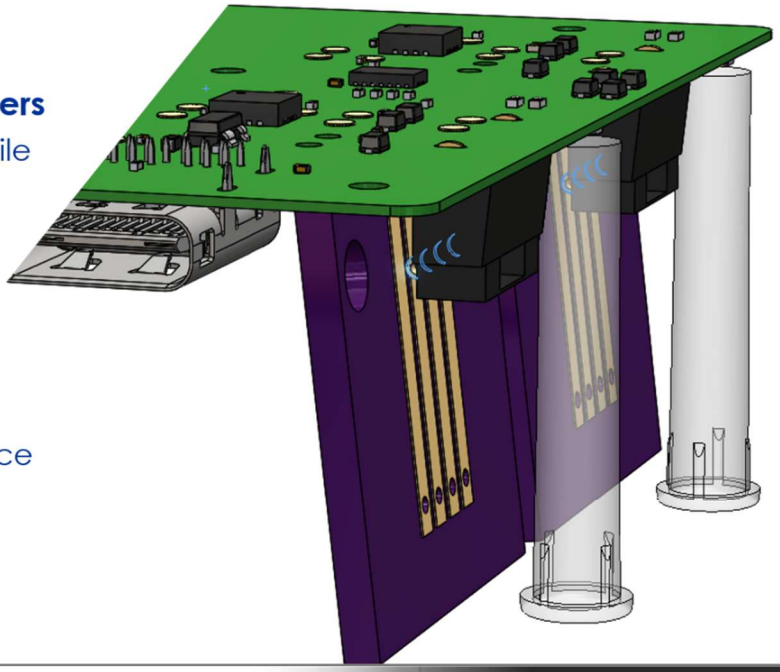
1. Alignment pin **reduces rotational error** by aligning with bushing.
2. Spring plunger presses into DUT to ensure it is fully seated.
3. Probes press into test points, making electrical contact.



Fixture Internals

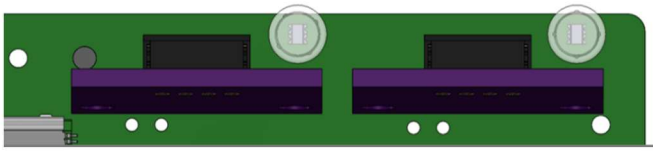
Fixed contact edges for battery spring fingers

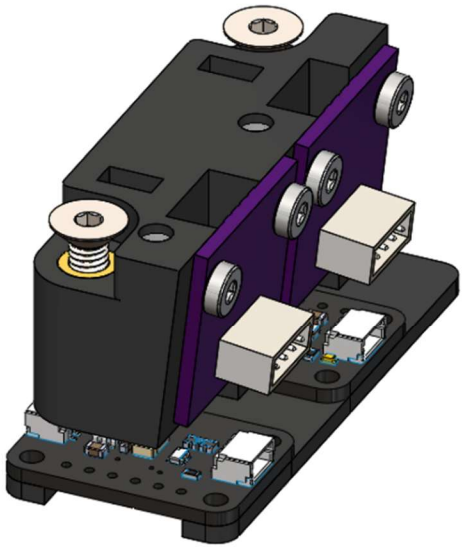
- Angle mitigates vertical shear risk on fragile J3/J4 fingers
- Eliminates moving parts



COTS light pipes

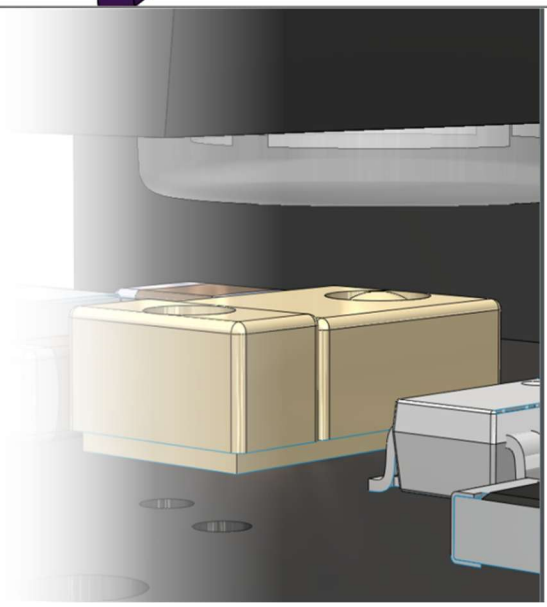
- Projects color onto internal diffused surface

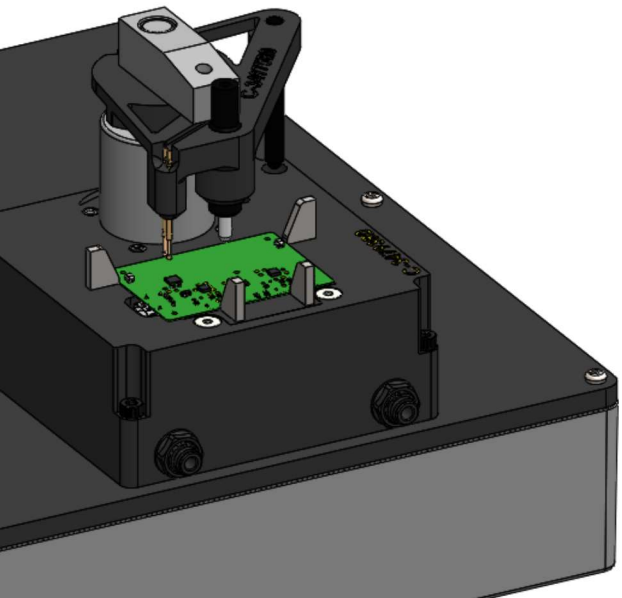


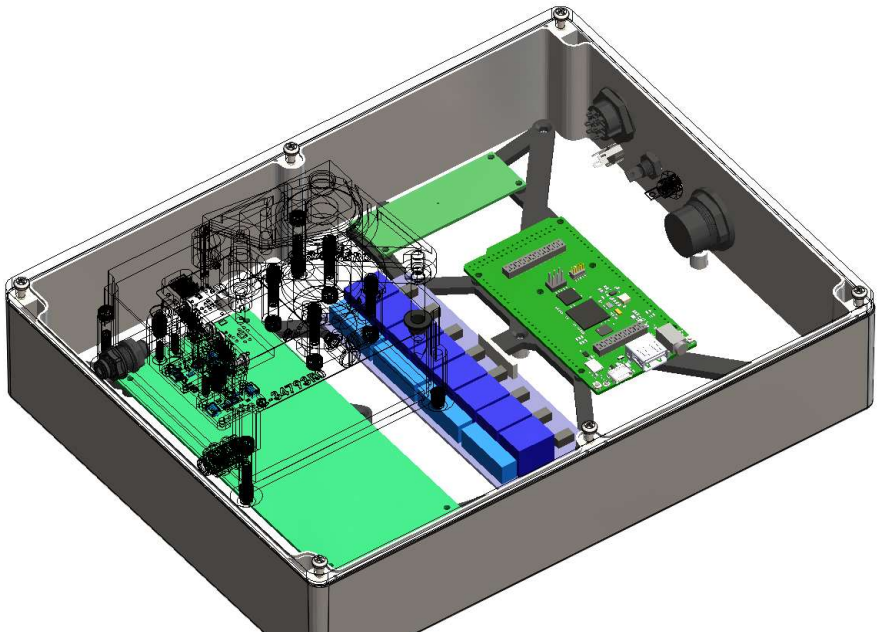


Interface Block

- Color sensors mount below
- Aligned to diffused surface







Test Cart & System Design

Wire Flags

Color-code

Red: High Voltage

Magenta: Wire deviates from the defined standard

Color Codes

Power Supply

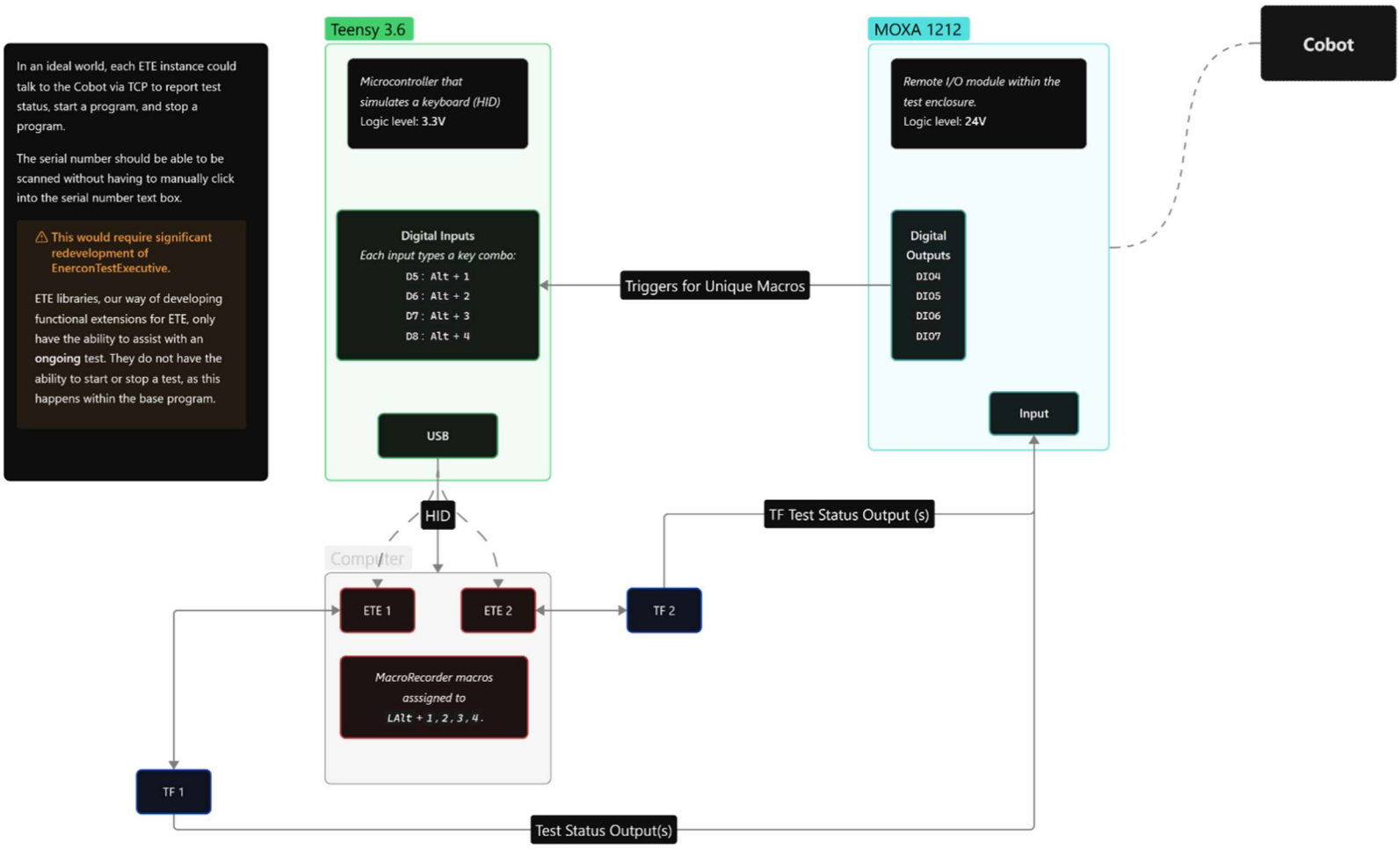
- +3.3V
- +24V
- 0V

Logic & Signals

- Cobot → Solenoids (24V)
- Fixture Status → Cobot (24V, from optocoupler)
- Cobot → Teensy (24V, to optocoupler)
- Cobot → Teensy (3.3V, from optocoupler)



System Communication Diagram *(Preliminary)*



IoT-Enabled Landscape Lighting *(Done Right)*

My neighbor wanted three landscape lighting zones within one backyard.

- **Patio**: Sunset to 9:00PM
  (For tea on warm summer nights)
- **Contour:** Sunset to Sunrise *(Edge lighting for visibility & safety)*
- **Highlights:** Sunset to 9:00 PM *(Uplighting for tall oak trees)*

Problem Statement

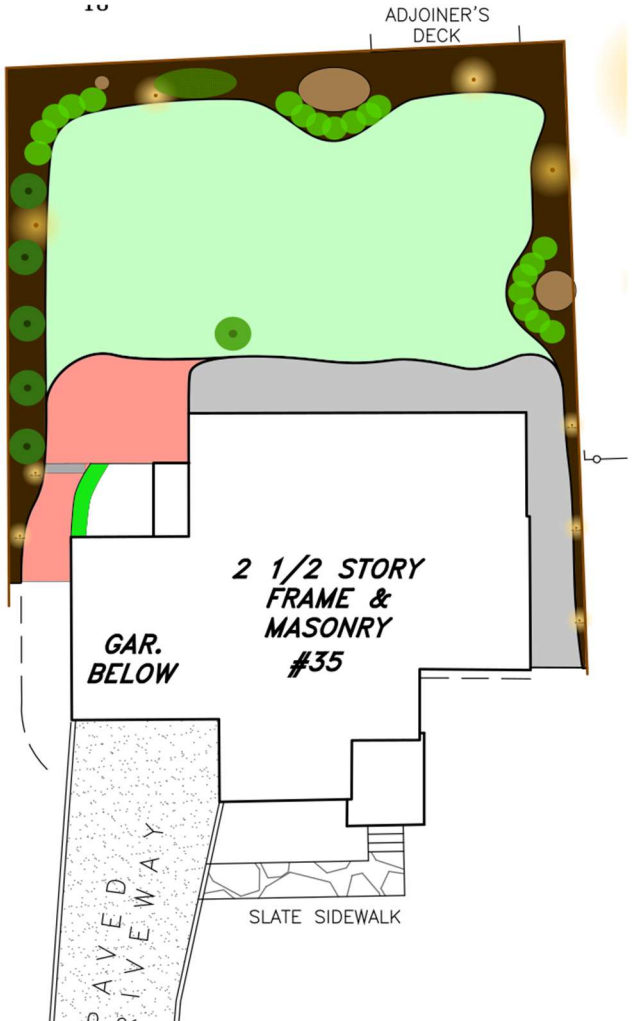
Commercial "smart landscape lighting" just isn't smart. You're provided with dated setups that switch entire transformers on and off, or "All-in-one" smart platforms that ask for service subscriptions to control the lights you own. **No thanks.**

Drafting the Installation

I drew out the existing landscape out over town property records with to-scale visuals of each lamp's rated lighting spread. This ensured that we didn't have "dead zones," After a few revisions, we solidified plans.

For smart zone control, this would be controlled using Home Assistant running on a Raspberry Pi over Z-Wave. The point is to keep the system locally controlled - without needing third party services.

"My lights should still turn on if the internet goes out."



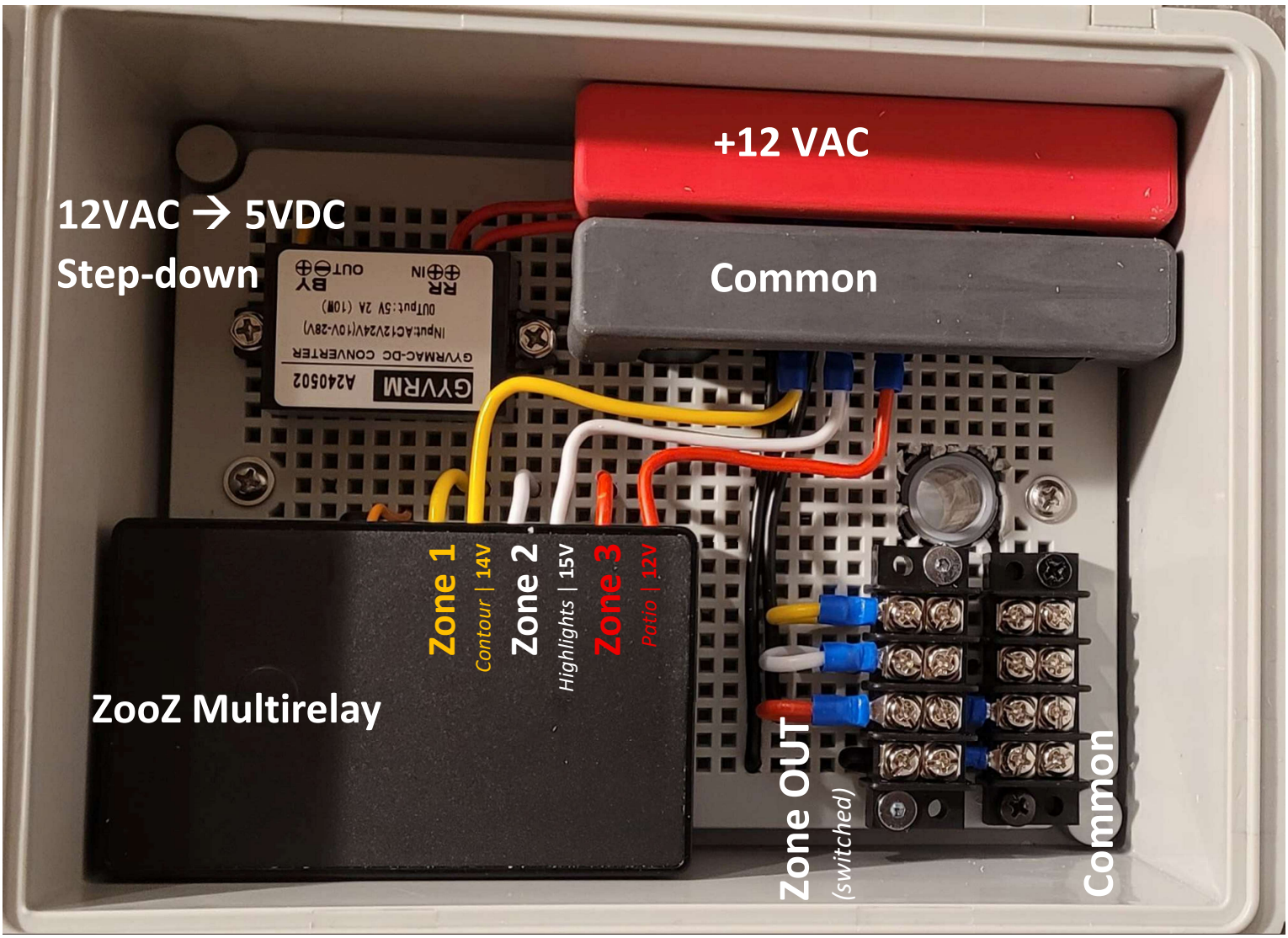


Control System

A custom control box sits outside to switch each zone. It features:

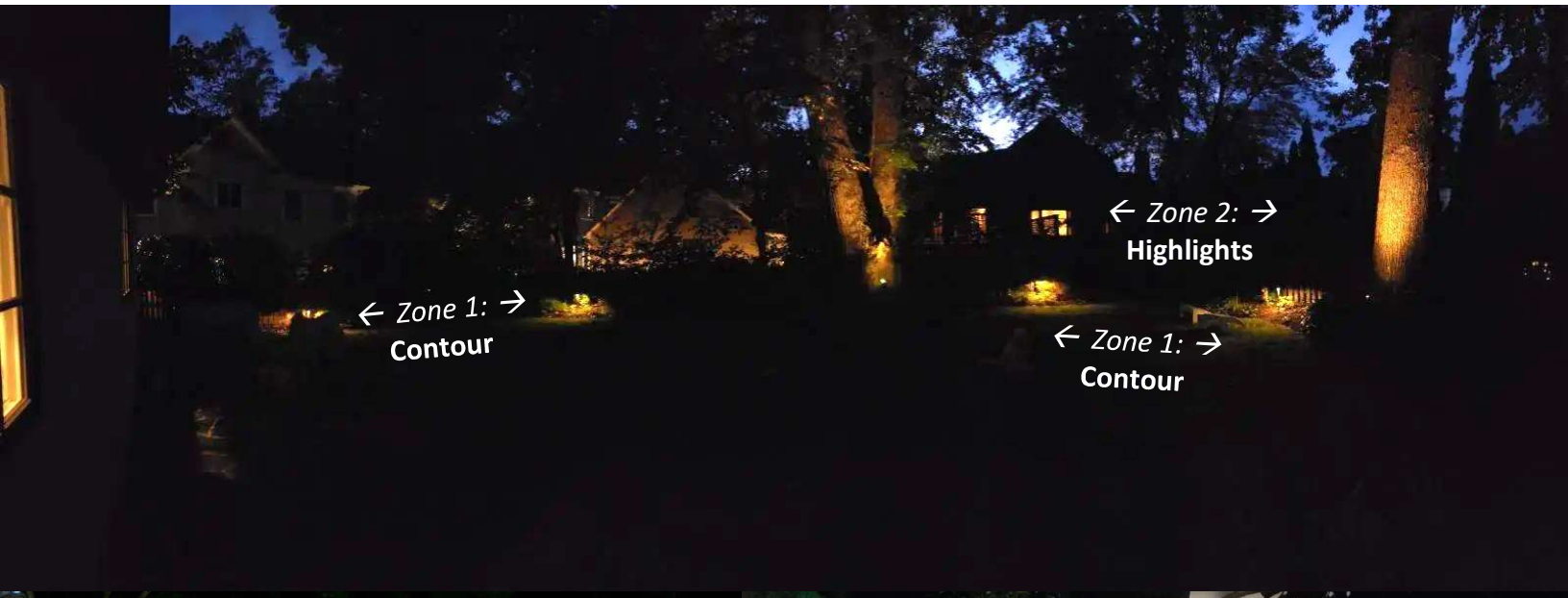
- Three Z-Wave connected relays
- Power distribution
- Three voltage levels (12V, 13V, 14V) to compensate for long-run voltage drop.

The control box is weatherstripped with Butyl. Inside, 20ft of soft conduit runs through the walls to the low-voltage transformer indoors. Outside, the conduit leads to a junction box buried in the mulch.



Project Gallery:

This project has a special place in my heart (and my back, ow.) I enjoy seeing automated installations integrate with day-to-day life – all while being independent from third-party services to turn your own lights on and off.







This step was notorious for tripping the homeowners at night as they walked around the house.

→ Unlike other timed zones, the **Contour** zone stays on from dusk to dawn and in rainy conditions for visibility.